DIBYAJYOTEE GHOSH

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PROFESSIONAL SUMMARY Results-driven Electronics and Communication Engineering (ECE) student specializing in the rapid prototyping and integration of hardware-software systems. Proven ability to architect, deploy, and troubleshoot complex Edge Computing, IoT, and AI applications. Expert in leveraging AI-assisted development to accelerate deployment timelines, architect physical circuits, and build full-stack embedded solutions. Seeking to leverage cross-disciplinary technical expertise in an innovative engineering environment.

EDUCATION Bachelor of Technology in Electronics and Communication Engineering
Institute of Engineering and Management, Kolkata *Expected Graduation: June 2028* |

CGPA: 8.83

TECHNICAL SKILLS

- **Programming (Basics & Applied):** Python (Scripting & AI-Assisted Development), C/C++ (Basic Arduino Logic), MATLAB (Familiar)
- **Hardware & Microcontrollers:** Raspberry Pi, Arduino Uno, Sensor Integration (MQ-Series Gas Sensors, Ultrasonic, Piezoelectric)
- **Applied Electronics:** Rapid Hardware Prototyping, Basic Circuit Troubleshooting, Component Wiring & Breadboarding
- **Tools & Methodologies:** AI-Assisted System Design, API Integration, Git/GitHub (Basics)
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ENGINEERING PROJECTS

Adaptive Hybrid Edge-Olfaction for Perishable Logistics

- Architected a full-stack hardware-software monitoring solution for perishable goods logistics by integrating a high-density environmental sensor array (MQ-series, HC-SR04).
- Designed a dual-processor architecture utilizing Arduino for analog signal processing and Raspberry Pi 4 for primary edge computing and web serving.
- Implemented Edge Computing principles to enable real-time data processing and decision-making directly on the device, minimizing latency and eliminating cloud-hosting costs.

- Developed predictive models utilizing non-linear decay equations to calculate produce degradation risk and estimate remaining shelf life based on raw sensor telemetry.
- Engineered the system for robust, low-power operation, demonstrating expertise in practical IoT deployment and complex sensor integration.

PoultryHealth-AI (Acoustic Disease Detector)

- Deployed a lightweight acoustic AI machine learning model on a Raspberry Pi 3B for agricultural monitoring.
- Utilized a pre-trained YAMNet architecture and custom MFCC edge models to accurately detect poultry respiratory anomalies from environmental audio feeds.

FC Manager AI

- Developed a Python-powered tactical prediction engine for football simulations using Reinforcement Learning modules.
- Integrated Optical Character Recognition (OCR) for automated squad detection, analyzing meta-mechanics and player traits to generate optimized tactical instructions.

CERTIFICATIONS

- **Cyber Security Fundamentals:** University of London
- **Generative Artificial Intelligence (GenAI):** 0x.Day & UEM
- **Intro to Information Technology & AWS Cloud:** Amazon Web Services
- **Introduction to Programming with MATLAB:** Vanderbilt University

EXTRACURRICULAR ACTIVITIES & INTERESTS

- **Active Volunteer, Binapani Square Saraswati Puja:** Assisted senior committee members with event setup, community coordination, and general logistics for a large-scale neighborhood cultural celebration.
- **Personal Interests:** Lifelong football player and tactical enthusiast; practiced in traditional sketching and drawing.